

An Exact Finite-Size Bulk–Edge and Quench Atlas for the SSH Chain

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Abstract

For the balanced open Su–Schrieffer–Heeger chain of M cells, we derive the full characteristic polynomial and prove that a strictly hyperbolic edge pair exists if and only if $w/v > (M+1)/M$. We give its exact splitting, two-ended eigenvectors, decay bound, critical linear taper, and fixed-ratio asymptotic.

1 Finite hyperbolic edge threshold

Let $M \geq 2$ and $v, w \geq 0$. In the ordered basis $(A_1, \dots, A_M, B_1, \dots, B_M)$, the open Hamiltonian is

$$H_O = \begin{pmatrix} 0 & T \\ T^* & 0 \end{pmatrix}, \quad T = vI + wS_-, \quad (S_-)_{j,j-1} = 1. \quad (1)$$

Thus $\Gamma = \text{diag}(I, -I)$ obeys $\Gamma H_O \Gamma = -H_O$.

Theorem 1 (Characteristic and sharp finite edge theorem). *Put $q_M(y) = \det(yI - TT^*)$. Then*

$$q_0 = 1, \quad q_1 = y - v^2, \quad q_m = (y - v^2 - w^2)q_{m-1} - v^2w^2q_{m-2}. \quad (2)$$

For $v, w > 0$, set $x = (E^2 - v^2 - w^2)/(2vw)$. With U_m the Chebyshev polynomial of the second kind,

$$\det(EI - H_O) = (vw)^M \left[U_M(x) + \frac{w}{v} U_{M-1}(x) \right]. \quad (3)$$

There is exactly one secular root $x = -\cosh \kappa < -1$, and hence exactly one eigenvalue pair $\pm E_e$ outside the trigonometric sector, if and only if

$$\frac{w}{v} > \frac{M+1}{M}. \quad (4)$$

For the unique $\kappa > 0$ determined by

$$\frac{w}{v} = \frac{\sinh((M+1)\kappa)}{\sinh(M\kappa)}, \quad (5)$$

the positive energy and one unnormalised positive-energy vector are

$$E_e = v \frac{\sinh \kappa}{\sinh(M\kappa)} = w \frac{\sinh \kappa}{\sinh((M+1)\kappa)}, \quad (6)$$

$$a_j = (-1)^{j-1} \sinh((M+1-j)\kappa), \quad b_j = (-1)^{j-1} \sinh(j\kappa). \quad (7)$$

Thus (a, b) has energy E_e and $(a, -b)$ has energy $-E_e$. The profiles decay strictly inward:

$$\frac{|a_{j+1}|}{|a_j|} < e^{-\kappa}, \quad \frac{|b_j|}{|b_{j+1}|} < e^{-\kappa} \quad (1 \leq j < M). \quad (8)$$

At equality in (4), $x = -1$ and $E = v/M$. After common division of a, b by κ (equivalently, after overall normalisation), the $\kappa \downarrow 0$ profiles are $a_j = (-1)^{j-1}(M+1-j)$ and $b_j = (-1)^{j-1}j$.

Proof. The leading diagonal entry of TT^* is v^2 ; every subsequent one is $v^2 + w^2$, and the off-diagonal entries are vw . Its continuant is (2). Normalising this recurrence gives (3); in particular, the factor $(vw)^M$ cannot be dropped. The M roots are real and simple because TT^* is an irreducible Jacobi matrix. Also $\det T = v^M$, so $E = 0$ is impossible when $v > 0$.

For $x = -\cosh \kappa$, the identity $U_m(-\cosh \kappa) = (-1)^m \sinh((m+1)\kappa)/\sinh \kappa$ reduces the secular equation to (5). Its logarithmic derivative is

$$(M+1) \coth((M+1)\kappa) - M \coth(M\kappa) > 0,$$

because $s \mapsto s \coth(s\kappa)$ is strictly increasing. The ratio therefore rises from $(M+1)/M$ to infinity. No root lies above $x = 1$, where both Chebyshev terms are positive, so this proves the iff statement and uniqueness. Direct hyperbolic addition verifies $Tb = E_e a$ and $T^*a = E_e b$, including both expressions in (6). Finally,

$$\frac{\sinh(n\kappa)}{\sinh((n+1)\kappa)} = e^{-\kappa} \frac{1 - e^{-2n\kappa}}{1 - e^{-2(n+1)\kappa}} < e^{-\kappa},$$

which proves (8). Taking $\kappa \downarrow 0$ gives the threshold energy; applying the common $1/\kappa$ rescaling before that limit gives the nonzero tapers. $\square$

Corollary 2 (Finite splitting). *For fixed $r = w/v > 1$ and all sufficiently large M , the hyperbolic root exists, $\kappa_M \rightarrow \log r$, and*

$$E_e \sim w(1 - r^{-2})r^{-M}. \quad (9)$$

Consequently the finite pair is never exactly zero for $v > 0$, although its splitting is exponentially small.

Proof. Writing $z = e^{-\kappa}$ turns (5) into $r = z^{-1}(1 - z^{2M+2})/(1 - z^{2M})$, hence $z \rightarrow r^{-1}$. Equation (6) becomes $E_e = vz^{M-1}(1 - z^2)/(1 - z^{2M})$, which gives (9). $\square$

Source lineage

References

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- [3] J. K. Asbóth, L. Oroszlány, and A. Pályi, “The Su–Schrieffer–Heeger Model,” in *A Short Course on Topological Insulators* (Springer, 2016). DOI: [10.1007/978-3-319-25607-8_1](https://doi.org/10.1007/978-3-319-25607-8_1).